Local Food Wastage Management System
Business Insights & Recommendations Report

1. Project Summary

This project built a data-driven system to track, analyze, and reduce local food wastage
by connecting food providers (restaurants, grocery stores, farms) with receivers (NGOs, shelters, individuals) across multiple cities.

Dataset Size:
- 1,000 Providers
- 1,000 Receivers
- 1,000 Food Listings
- 1,000 Claims

2. Key Business Insights

Insight 1: Claim Rate Reflects System Efficiency
The overall claim rate (total claims / total listings × 100) directly measures how effectively the system is redistributing food. A claim rate below 70% indicates significant food going unclaimed and likely wasted.

Recommendation: Set a target claim rate of 85%+ and monitor it monthly using Query Q1 as a KPI dashboard metric.

Insight 2: Provider Type Drives Donation Volume
Certain provider types (e.g. restaurants, grocery stores) consistently donate higher quantities than others (e.g. individual households). This concentration means that losing even a few high-volume providers significantly impacts supply.

Recommendation: Prioritize retention and engagement programs for top provider types. Introduce a provider loyalty/recognition program to keep high-volume donors active.

Insight 3: Food Expiry is a Critical Wastage Driver
A significant portion of food listings expire before being claimed. Food expiring within 7 days needs urgent redistribution but often doesn't get claimed in time due to lack of visibility.

Recommendation:
- Add an automated alert system that notifies nearby receivers when food is expiring within 48 hours.
- Prioritize expiring-soon listings at the top of the receiver's feed.
- Track the "expired unclaimed" rate monthly as a wastage KPI.

Insight 4: Geographic Mismatch Between Supply and Demand
Some cities have high food listing volumes but low claim rates, while others have high receiver counts but limited listings. This supply-demand mismatch is a primary cause of wastage.

Recommendation:
- Use Query Q5 (city-wise availability) and Q13 (receivers by city) together to identify mismatch cities.
- Run targeted provider recruitment campaigns in cities with high receiver counts but low supply.
- Consider cross-city food transfer partnerships for surplus cities.

Insight 5: Certain Food Types Have Low Claim Rates
Not all food types are claimed equally. Perishable items like dairy and cooked meals have lower claim rates than dry goods and packaged foods due to shorter shelf life and transport challenges.

Recommendation:
- For low-claim food types, reduce the listing-to-claim window (e.g. prioritize same-day claims for cooked meals).
- Partner with receivers who specialize in handling perishable items (e.g. soup kitchens, daily meal programs).

Insight 6: Monthly Trends Reveal Seasonal Patterns
Claim volumes fluctuate month-to-month. Peaks may align with festivals, harvest seasons, or community events. Troughs may indicate operational gaps or low receiver engagement.

Recommendation:
- Use monthly trend data (Query Q9) to pre-plan capacity during high-volume months.
- Run receiver activation campaigns during low-claim months to maintain consistent utilization.

Insight 7: Inactive Providers Represent a Lost Opportunity

A subset of providers have listed food that was never claimed (Query Q16). This discourages them from listing again, reducing overall supply.

Recommendation:
- Reach out to inactive providers with personalized outreach explaining why their listings went unclaimed.
- Match inactive providers with nearby receivers proactively rather than waiting for receivers to discover listings.

Insight 8: Top 10 Wastage Items Need Targeted Action
Query Q14 identifies specific food items with the highest unclaimed quantities. These items represent the biggest opportunity for wastage reduction.

Recommendation:
- Create dedicated campaigns for top wastage items (e.g. "Today's Surplus: Rice available in your area").
- Work with NGOs to create standing orders for commonly wasted bulk items.

3. Summary of Recommendations

Priority	Recommendation	Impact
High	Expiry alert system for 48-hour listings	Reduce expired unclaimed food
High	Target mismatch cities for provider recruitment	Balance supply-demand
High	Monthly KPI dashboard tracking claim rate	Measure system efficiency
Medium	Loyalty program for top providers	Retain high-volume donors
Medium	Receiver activation campaigns in low months	Maintain consistent claims
Medium	Perishable food fast-track claiming	Reduce cooked meal wastage
Low	Cross-city food transfer partnerships	Utilize surplus in other cities
Low	Inactive provider outreach program	Re-engage lapsed donors

4. Technical Achievements

Component	Details
Database	PostgreSQL with 4 normalized tables and foreign key relationships
Data Pipeline	Python ETL script loading 4,000 records with null handling
SQL Analysis	17 analytical queries covering trends, wastage, and performance
EDA	13 visualizations across univariate, bivariate, and multivariate analysis
Application	Full Streamlit app with 5 pages: Home, Explorer, SQL, Charts, CRUD
Data Quality	Null handling, type conversion, FK validation during data loading

5. Future Scope

Real-time notifications: WhatsApp/email alerts for expiring food
ML model: Predict which listings are likely to go unclaimed
Mobile app: Allow providers to list food directly from their phone
Map view: Geographic visualization of supply vs demand by city
Deployment: Host on Streamlit Cloud or AWS for public access